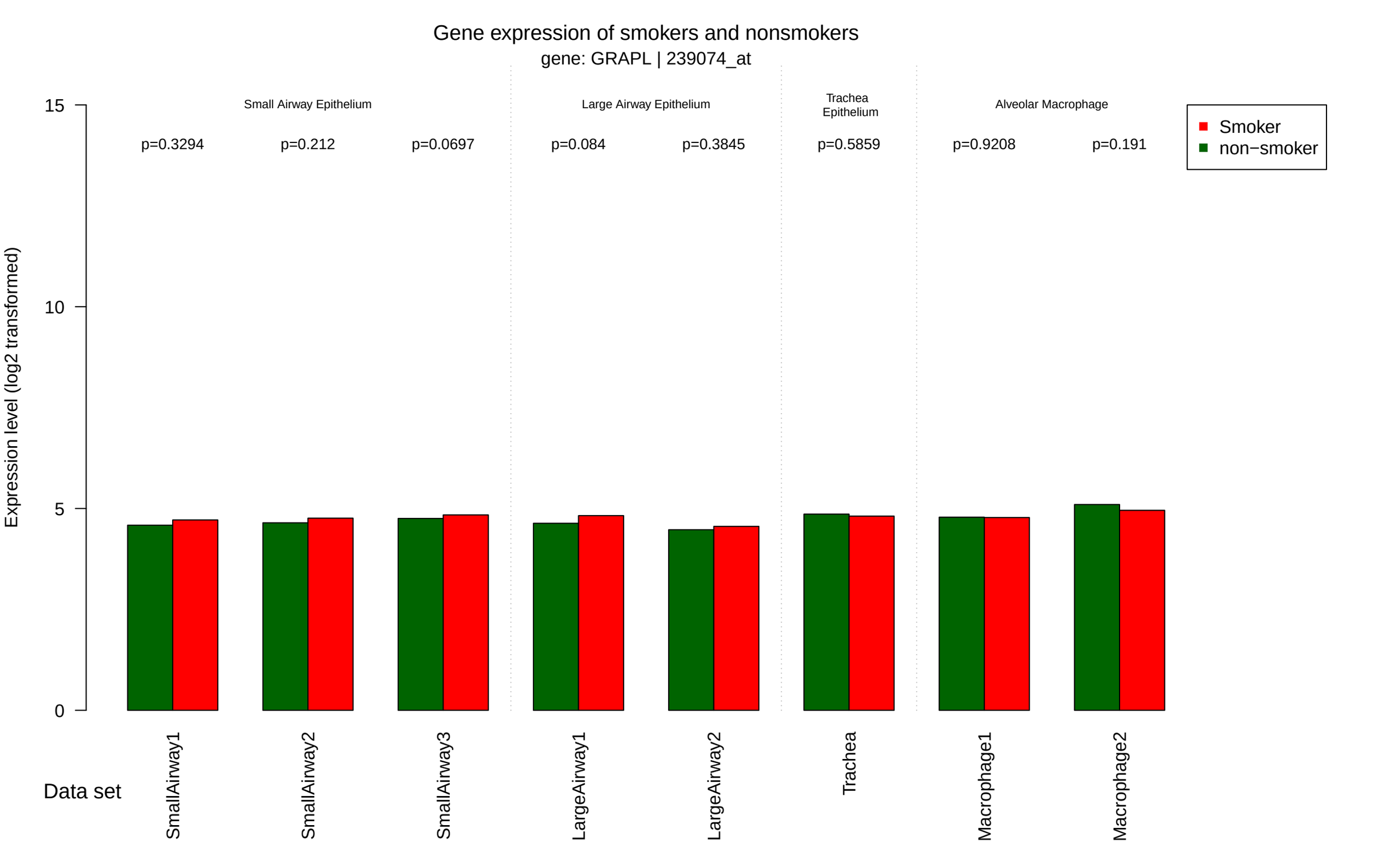
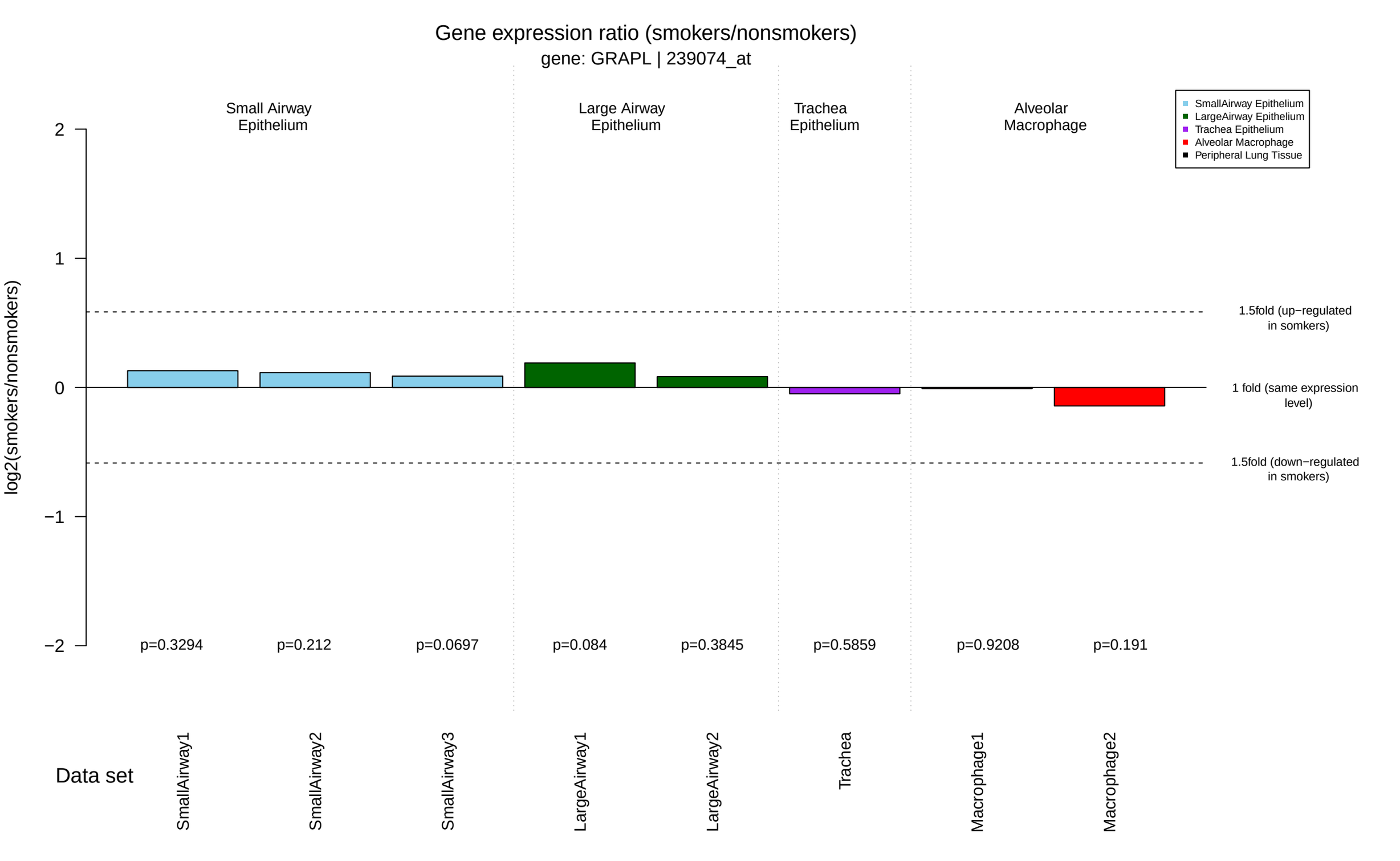
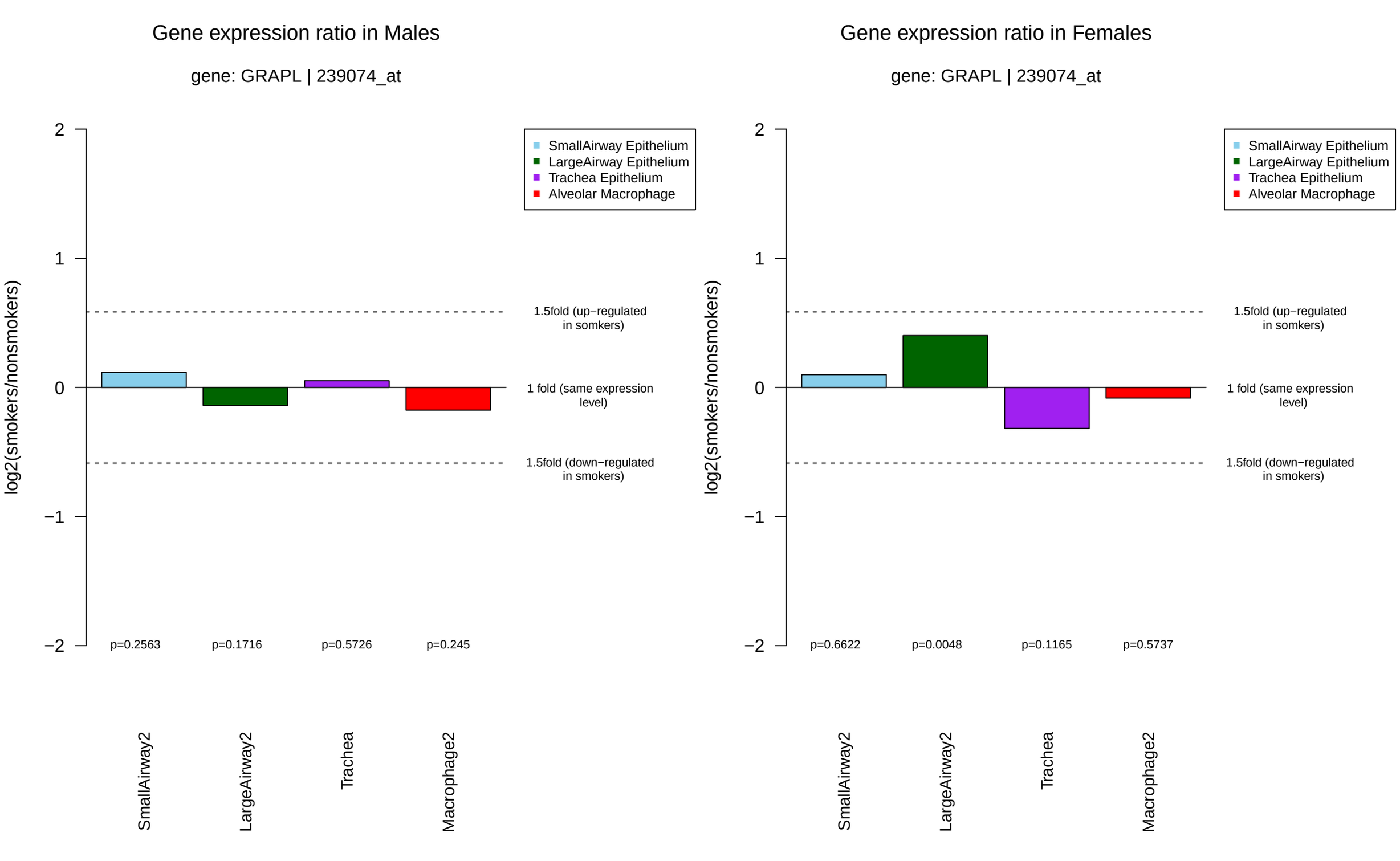
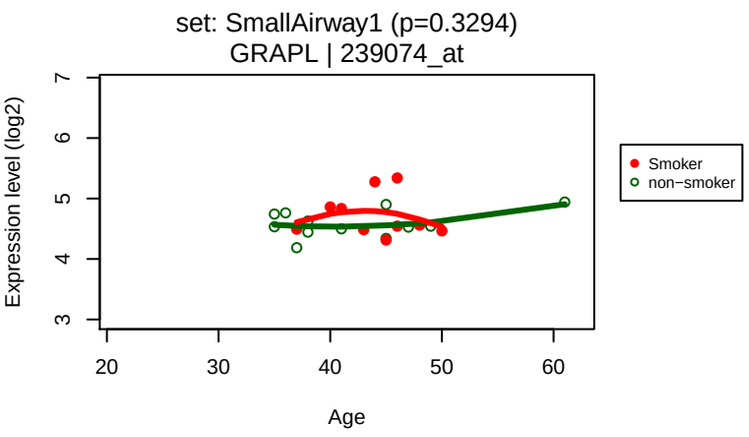
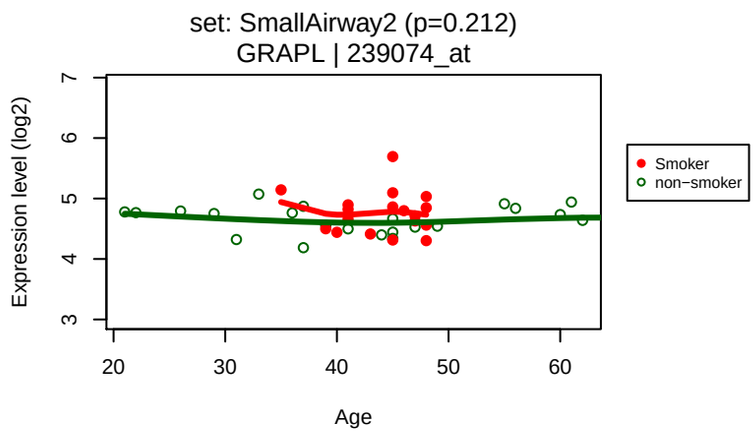
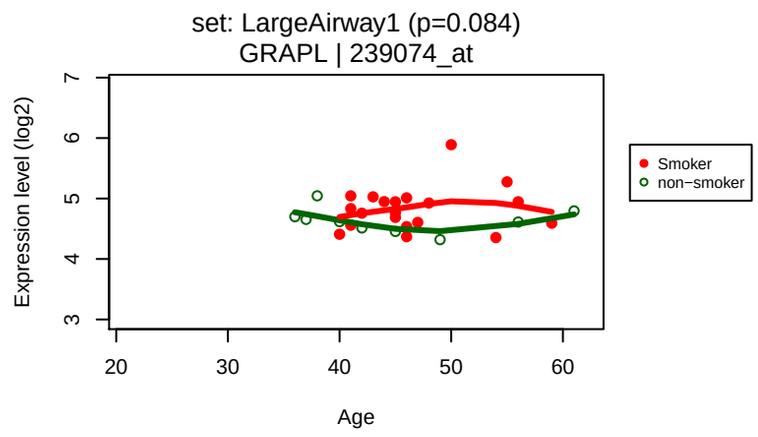

Gene expression of smokers and nonsmokers
gene: GRAPL | 239074_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3294
p=0.212
p=0.0697
p=0.084
p=0.3845
p=0.5859
p=0.9208
p=0.191
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

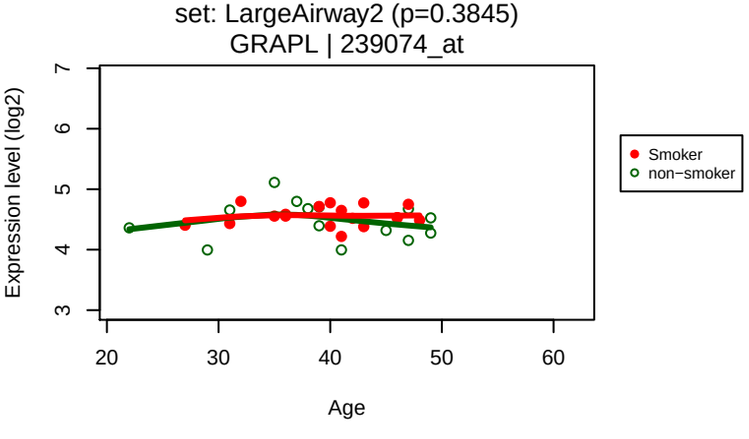

Gene expression ratio (smokers/nonsmokers)
gene: GRAPL | 239074_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3294
p=0.212
p=0.0697
p=0.084
p=0.3845
p=0.5859
p=0.9208
p=0.191
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

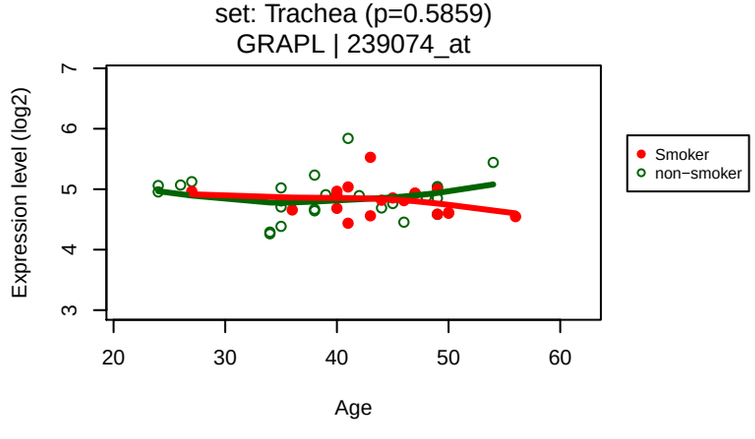

Gene expression ratio in Males
gene: GRAPL | 239074_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2563
p=0.1716
p=0.5726
p=0.245
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRAPL | 239074_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6622
p=0.0048
p=0.1165
p=0.5737
SmallAirway2
LargeAirway2
Trachea
Macrophage2

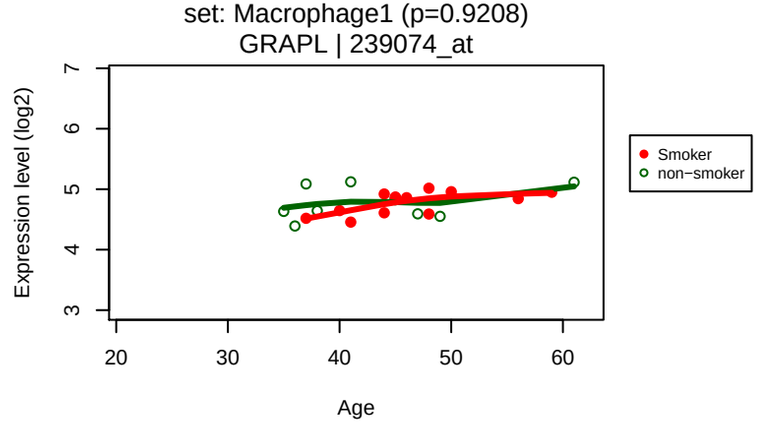

set: SmallAirway1 (p=0.3294)
GRAPL | 239074_at
Expression level (log2)
Age
Smoker
non−smoker

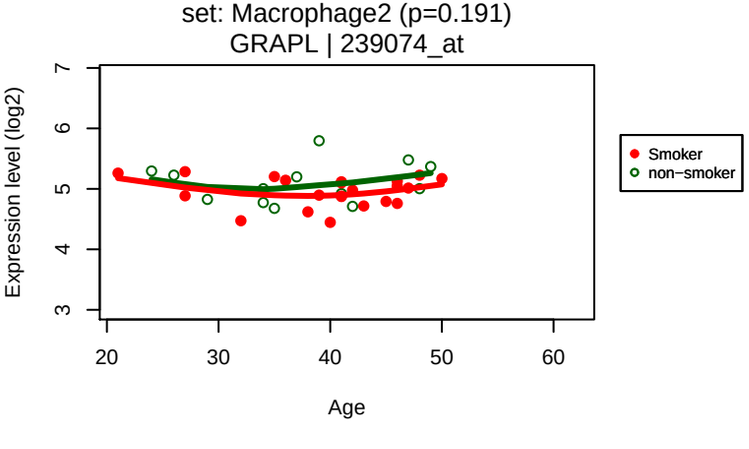

set: SmallAirway2 (p=0.212)
GRAPL | 239074_at
Expression level (log2)
Age
Smoker
non−smoker

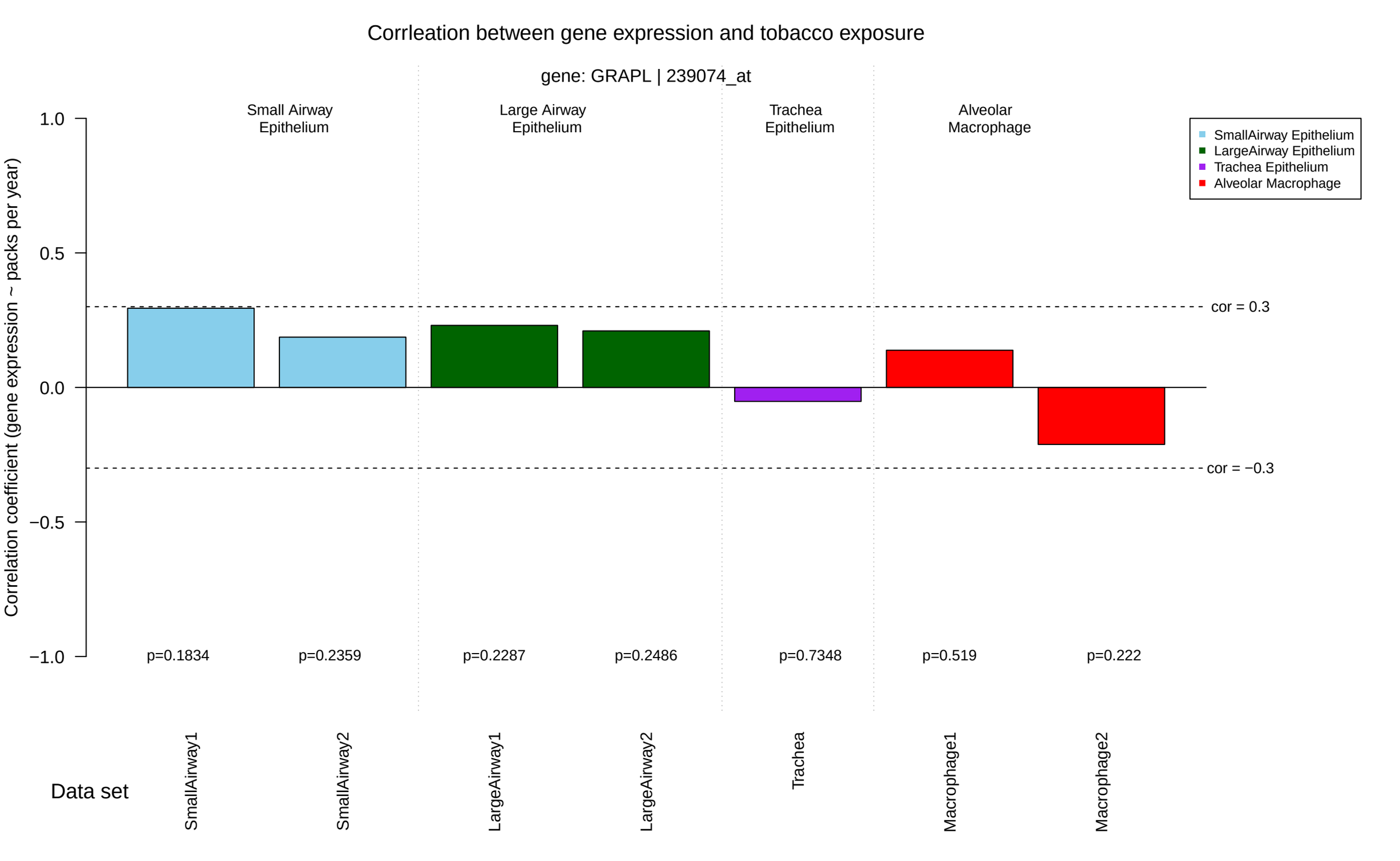

set: LargeAirway1 (p=0.084)
GRAPL | 239074_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3845)
GRAPL | 239074_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5859)
GRAPL | 239074_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9208)
GRAPL | 239074_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.191)
GRAPL | 239074_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRAPL | 239074_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1834
p=0.2359
p=0.2287
p=0.2486
p=0.7348
p=0.519
p=0.222
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2